Evaluation of Patient Oriented Drug to Drug Interactions Information Systems

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Introduction

- Adverse Drug Reactions (ADR) contribute significantly to medical errors.
- ADR is a serious concern as it resulted in over 325,000 serious health outcomes and over 54,000 deaths in the year 2012 (FDA, 2013).
- Drug-Drug Interactions (DDI) account for 9-70% of all Adverse Drug Reactions (Jankel, CA et al., 1990).
- Accurate reporting of DDI is essential for preventative care.
- There is a lack of studies that evaluate the performance of DDI Information Systems in Patient-Oriented settings.



Figure 1. Drug interactions icon adapted from: http://www.pepid.com/drug-interactions/

Research Questions

- When looking at patient oriented DDI Checkers, can the websites correctly identify and indicate severity related to the clinically significant drug pairs?
- Are the DDI Checkers Patient-Oriented (easy access and provide easily understood information)?

Methodology

- The study analyzed public drug consumer websites and determined if currently available information are accurate in regards to clinically significant DDI.
- 44 of 65 Websites were included in the study by meeting the inclusion/exclusion criteria.
- Severity Rating differences documented individually based on information of DDI presence.
- DDI Checkers were graded on severity ratings, Information Capacity Scores, Patient Usability Scores, and Flesch-Kincaid grading.

Information Capacity Score

DDI	Severity Rating	Drug-Food Reaction	Drug-Alcohol Reaction	Therapeutic Duplication	Max Score
0-5	0-5	0-5	0-4	0-2	21

Patient Usability Score

Alert Icons	Color Coding	Severity Rating Scale	Medication Pick List	Login/Profile Save for Multiple Drugs	Total Score
0-1	0-1	0-1	0-1	0-1	0-5

Figure 2. Quantification of the websites' ability to provide relevant and usable information was assessed through an analysis of the Information Capacity Score, Patient Usability Score.

Results

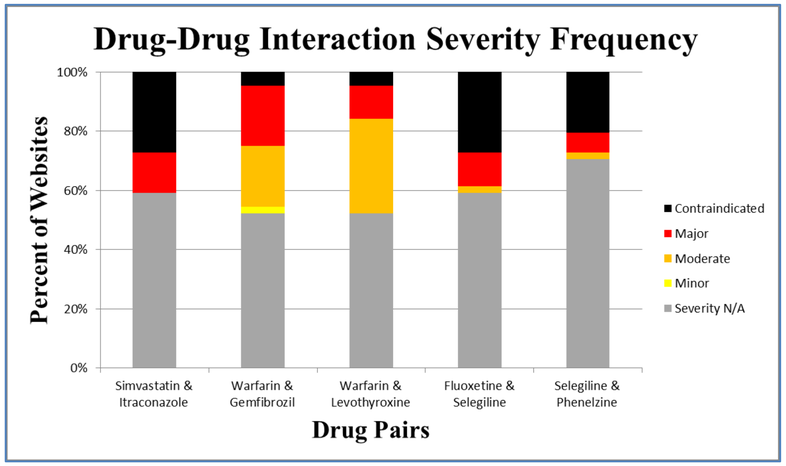


Figure 3. Severity level variation that existed between sites for each DDI pair (% of websites).

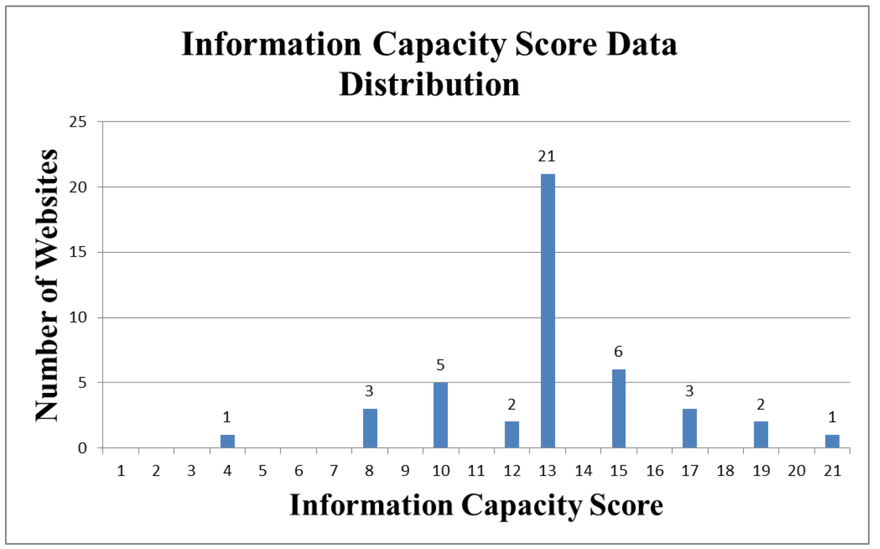


Figure 4. Information Capacity Score distribution graph.

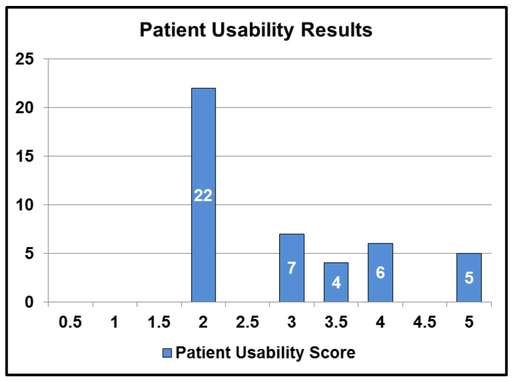


Figure 5. Displays the number and distribution of websites' user friendliness ratings.



Negative Drug Interaction

Figure 6. Drug interaction in action adapted from: http://www.pinterest.com/pin/4638 7083033 9409213/

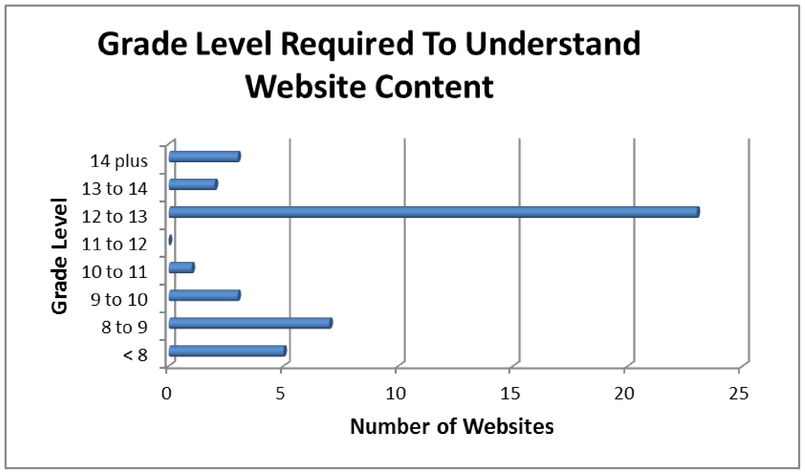


Figure 7. Displays the reading level for DDI content on each site (Flesch-Kincaid Grading model).

Discussion

- Majority of Patient-Oriented DDI Information Systems operate similarly.
- Many websites containing DDI checkers identified an existing drug interaction, but failed to provide a severity rating for the drug pairs studied.
- Of those that did have a severity rating, discrepancies existed between the level of severity reported for the five drug pairs.
- Almost half of the websites had identical scores in all categories.
- **Information Capacity Scores and Patient Usability Scores for websites were frequently quite low, raising concerns regarding information quality and website usability.**

Implications

- While there was consistency in how DDI Information Systems operate across websites, they were similar deficiencies in providing consumer-friendly information.
- The majority of websites may not be Patient-Oriented with high Flesch-Kincaid grade levels and low Patient Usability Scores.
- Discrepancies between severity ratings may cause consumers to distrust available DDI information.
- This research will provide important preliminary data for future assessments of Patient-Oriented DDI Information Systems.

References

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- Saverno, Kim, Lisa Hines, and Terri Warholak. "Ability of Pharmacy Clinical-Decision Support software to alert users about clinically important drug-drug interactions." *Journal of the American Medical Informatics Association* 18 (2011): 32-37.